NABIHAH BINTI TAQUDDIN AZMY

Software Engineer | Embedded & Full-Stack Systems

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PROFESSIONAL SUMMARY

Skilled Software Engineer and Technical Lead with over 5 years of professional experience building full-stack web applications, relational database infrastructure, and leading software platform delivery. Combines a strong software engineering foundation (Node.js, Express, NestJS, RESTful APIs, MySQL, MongoDB) with hands-on capabilities in embedded hardware integration (ESP32-S3 microcontrollers, Embedded Touchscreen GUIs, BLE telemetry, Edge AI). Experienced lead Person-in-Charge (PIC) for multi-module enterprise platforms, managing end-to-end SDLC workflows, Functional Specification Documentation (FSD), system monitoring, and cross-functional project execution.

CORE COMPETENCIES & TECH STACK

- **System Architecture & Integration:** End-to-End System Design, Hardware-Software Integration, Embedded Touchscreen GUI Architecture, RESTful API Design, System Process Mapping, FSD Authoring & Technical Documentation.
- **IDEs, Embedded UI & Hardware Frameworks:** ESP-IDF (C/C++), Arduino IDE, SquareLine Studio (LVGL Framework), Microcontrollers (ESP32-S3, Arduino), BLE Telemetry & Protocols, Sensor Data Acquisition, Edge AI Applications (Edge Impulse Studio).
- **Backend & Software Development:** Node.js, NestJS, Express.js, JavaScript, Python, C / C++, PHP (CodeIgniter / Laravel), RESTful Web Services.
- **AI Tooling & Developer Productivity:** ChatGPT, Gemini, Antigravity (AI-Assisted Code Generation, Rapid Prototyping & Workflow Acceleration).
- **Databases & DevOps Tools:** MySQL, MongoDB, Relational Database Design, AWS S3, Postman, Git / GitLab, VS Code.

PROFESSIONAL EXPERIENCE

Assistant Project Manager cum Software Engineer

Ingenious Technology Solutions Sdn. Bhd., Perai, Penang | Dec 2022 – Present

- **Enterprise Platform Leadership:** Served as primary technical PIC for a 14-module enterprise platform (MECHIS), monitoring system health and coordinating with IT infrastructure teams to execute database expansions, safeguarding 24/7 uptime.
- **Embedded Touchscreen GUI & Hardware Integration:** Designed Embedded Touchscreen GUI for ESP32-S3 AMOLED display via SquareLine Studio (LVGL), integrating touch UI with firmware logic and sensor data streaming pipelines.
- **Hardware-Software IoT Data Pipelines:** Developed end-to-end telemetry workflows pairing edge MCUs (ESP32 / Arduino) with software interfaces via BLE, enabling real-time sensor data streaming and monitoring.
- **Embedded AI & Edge Application:** Applied Edge AI models (Edge Impulse) for acoustic and signal classification, evaluating performance, signal processing, and low-power hardware integration.
- **Internal Tooling & System Automation:** Architected and deployed a production-ready, role-based Ticketing Support System (Node.js/Express, MySQL) to automate incident management workflows and enforce SLA resolution timelines.
- **Python Data Diagnostics & AI Orchestration:** Utilized Python scripts and AI-assisted developer workflows for serial telemetry logging, signal data analysis, and rapid system integration testing.
- **SDLC Governance & Requirement Analysis:** Authored Functional Specification Documentation (FSD), evaluated Change Requests (CRs), and directed UAT cycles and client onboarding to maintain system stability.

Software Engineer

Airei Sdn. Bhd., Cyberjaya, Selangor | Apr 2020 – Nov 2022

- **Application Development & Support:** Executed full SDLC development for web applications using PHP (CodeIgniter) and JavaScript, adhering to strict client SLAs and system specifications.
- **Quality Assurance & Documentation:** Conducted functional testing, authored technical end-user documentation, and supported production release deployments.

Software Developer Intern

Majlis Bandaraya Ipoh, Perak | Feb 2019 – Jul 2019

- **Web Platform Development:** Designed and built a web-based meeting scheduling platform (eJadual) using JavaScript, AJAX, and relational databases to automate municipal scheduling workflows.

EDUCATION

Bachelor's Degree in Software Engineering | Universiti Putra Malaysia (UPM) | 2015 – 2020

- **Final Year Project:** Two-Factor Authentication (2FA) Security Implementation for Internal Document Management Systems (Laravel / REST API).

CERTIFICATIONS & CONTINUOUS LEARNING

- **Edge AI Fundamentals | Edge Impulse (Aug 2026):** Applied study in Edge AI architecture, signal processing, and deploying machine learning models on resource-constrained microcontrollers.
- **NestJS Enterprise Application & REST API Architecture | CodeSignal (Jan 2026):** Structured backend application architecture, dependency injection, REST API development, and MongoDB integration.
- **Backend Development with Node.js & Express.js | Codefinity (Oct 2025):** Server-side application logic, asynchronous routing, and database integrations.